

Each question is worth 5 points for a total of 40 points. The Extra Credit problem can get you up to an additional 5 points. NO CALCULATORS allowed. If you use a calculator, the entire test will receive a score of 0.

Problem 1

Find the slope of the function $f(x) = x - 2x^2$ at the point $(1, -1)$. Then find an equation for the line tangent to the graph at the given point.

Problem 2

Find $\frac{dz}{dw}$ where $z = \frac{1}{\sqrt{w^2-1}}$.

Problem 3

Find the first and second derivatives of $y = \frac{x^3+7}{x}$ with respect to x .

Problem 4

Let $s = -t^3 + 3t^2 - 3t$ give the position of a body moving on a coordinate line, with s in meters and t in seconds over the interval $0 \leq t \leq 3$. Find (a) the body's displacement and average velocity over the interval, (b) the body's speed and acceleration at the end points of the interval, and (c) when, if ever, during the interval the body changes direction.

Problem 5

Find $\frac{dr}{d\theta}$ where $r = 4 - \theta^2 \sin(\theta)$.

Problem 6

Find the derivative of $h(x) = x \tan(2\sqrt{x}) + 7$ with respect to x .

Problem 7

Find the slope of $y^2 + x^2 = y^4 - 2x$ at the points $(-2, 1)$ and $(-2, -1)$.

Problem 8

If $x^2 + y^2 = 25$ and $dx/dt = -2$, then what is dy/dt when $x = 3$ and $y = -4$?

Extra Credit:

A 13-ft ladder is leaning against a house when its base starts to slide away. By the time the base is 12 ft from the house, the base is moving at the rate of 5 ft/sec. Find (a) how fast the top of the ladder is sliding down the wall then, (b) at what rate the area of the triangle formed by the ladder, wall, and ground is changing then, and (c) at what rate the angle between the ladder and the ground is changing then.